

Haolei Bai

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Work Experience

Westlake University , Hangzhou, China	Jan.2025 - Present
Research Internship, Worked on academic papers	
Agency for Science, Technology and Research (A*STAR) , Singapore	May.2024 - Nov.2024
Research Internship, Employed LLMs to extract events from unstructured documents	

Education

Nanyang Technological University , Singapore	Aug.2023 - Feb.2025
Master of Science in Signal Processing	
Qingdao University of Technology , Qingdao, China	Sep.2019 - Jun.2023
Bachelor of Engineering in Communication Engineering	

Publications

DICE: Diffusion Large Language Models Excel at Generating CUDA Kernels	Under Review
Haolei Bai , Lingcheng Kong, Xueyi Chen, Jianmian Wang, Zhiqiang Tao, Huan Wang	
- Curated a dataset tailored for CUDA kernels and proposed a new training strategy - Employed diffusion large language models to generate CUDA kernels	
ERC-SVD: Error-Controlled SVD for Large Language Model Compression	CPAL'26
Haolei Bai , Siyong Jian, Tuo Liang, Yu Yin, Huan Wang	
- Leveraged residual matrices to compensate for SVD, reducing the overall truncation loss - Proposed partial-layer compression strategy to reduce layer-wise error and mitigate error propagation	
Data Efficient Training for Egocentric Vision-based Action Recognition	Master's Thesis
Haolei Bai	
- Efficiently learn features with minimal labeled data using semi-supervised learning techniques - Applied multimodal learning frameworks, incorporating complementary data modalities to enhance model performance - Applied transformer-based architectures to analyze, implemented iterative experimentation and evaluation	
MobileKernelBench: Can LLMs Write Efficient Kernels for Mobile Devices?	DATA-FM @ ICLR'26
Xingze Zou, Jing Wang, Yuhua Zheng, Xueyi Chen, Haolei Bai , Lingcheng Kong, Syed A.R. Abu-Bakar, Zhaode Wang, Chengfei Lv, Haoji Hu, Huan Wang	
- Extended the scope of automated kernel generation to mobile domain - Proposed an evaluation framework called MobileKernelBench to evaluate LLM performance	

Scholarships

CFAR Internship Award for Research Excellence (CIARE)	2024
Excellent Student Scholarship	2019 - 2023

Skills & Languages

Skills: Python, Pytorch, Linux and C/C++
Languages: Chinese (Native), English (IELTS 6.5)